

SOCIAL MEDIA ENGAGEMENT ANALYSIS USING PYTHON

1. Project Overview

This project focuses on analyzing social media engagement data using Python to understand user interaction patterns, engagement trends, and factors influencing social media performance. The analysis includes data cleaning, exploratory data analysis (EDA), visualization, and generating insights to support data-driven decision-making.

2. Data Source

- **Dataset Name:** social_media_engagement_5000.csv
- **Source:** Dataset provided as part of the **Social Media Engagement Analysis Using Python** project (uploaded CSV file).
- **File Type:** Comma-Separated Values (.csv)
- **Number of Records:** 5,000
- **Number of Attributes:** 19

3. Problem Statement

This project aims to analyze the **Social Media Engagement** dataset using Python techniques such as data preprocessing, exploratory data analysis (EDA), statistical analysis, and data visualization. The objective is to identify key patterns, trends, and relationships among user demographics, post characteristics, and engagement metrics, thereby generating actionable insights to enhance social media performance and support better decision-making.

4. Objectives

- To understand the structure and characteristics of the social media engagement dataset.
- To clean and preprocess the dataset by handling missing values, duplicates, and inconsistent data.
- To perform Exploratory Data Analysis (EDA) to gain insights into user engagement patterns.
- To analyze key engagement metrics such as likes, comments, shares, impressions, and engagement rate.
- To identify trends and relationships between user demographics, post characteristics, and engagement levels.
- To create meaningful visualizations using Python libraries such as Matplotlib and Seaborn.
- To evaluate the impact of factors such as post type, sentiment, hashtags, device type, and follower count on engagement.
- To generate actionable insights that can help improve social media content strategies and audience engagement.
- To provide a foundation for future predictive analytics and data-driven decision-making.

5. Attribute (Column / Feature) Details

Attribute Name	Description
1. User_id	Unique identifier for each social media user.
2. Age	Age of the user.
3. Gender	Gender of the user.
4. Country	Country where the user is located.
5. Post_id	Unique identifier for each social media post.
6. Post_type	Type of content posted (e.g., Image, Video, Text, Reel).
7. Post_category	Category or topic of the post (e.g., Entertainment, Education, Sports).
8. Likes	Total number of likes received on the post.
9. Comments	Total number of comments received on the post.

10. Shares	Total number of times the post was shared.
11. Watch_time_sec	Total watch time of the post in seconds (mainly for video content).
12. Impression_count	Number of times the post was displayed to users.
13. Posted_at	Date and time when the post was published.
14. Follower_count	Number of followers of the user who created the post.
15. Is_verified	Indicates whether the user's account is verified (Yes/No or True/False).
16. Device_type	Device used to access or upload the content (e.g., Android, iOS, Web).
17. Sentiment	Sentiment associated with the post or audience reaction (Positive, Neutral, Negative).
18. Hashtags	Hashtags used in the post.
19. Engagement_rate	Overall engagement rate calculated based on user interactions such as likes, comments, and shares.

6. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

7. Data Pre-Processing and Feature Engineering

The dataset was prepared before analysis to improve data quality and ensure reliable results.

Data Pre-Processing Steps

- Imported the dataset into a Pandas DataFrame.
- Checked the dataset structure using head(), info(), and describe().
- Identified missing values using isnull().sum().
- Checked for duplicate records and removed them if present.
- Converted data into appropriate data types where necessary.
- Verified the dataset for consistency and accuracy before analysis.

Feature Engineering

- Created derived metrics where required for better analysis.
- Categorized numerical variables into meaningful groups when applicable.
- Prepared features for visualization and statistical analysis.
- Organized categorical variables for easier interpretation.

8. Analysis and Visualizations

The project analyzed the social media engagement dataset using exploratory techniques and visual representations.

The analysis included:

- Understanding data distribution.
- Comparing engagement across different categories.
- Identifying trends and patterns.
- Detecting relationships among variables.
- Finding possible outliers.

Visualizations were created using:

- Scatter
- Line Chart
- Bar Chart

- Pie Chart
- Histogram
- Box Plot
- Count Plot
- Bar plot
- Heatmap
- Plotly Interactive Chart

9. Exploratory Data Analysis (EDA)

EDA was performed to understand the characteristics of the dataset before detailed analysis.

The following activities were carried out:

- Examined dataset dimensions.
- Reviewed summary statistics.
- Identified missing values.
- Checked duplicate records.
- Analyzed numerical feature distributions.
- Explored categorical variable frequencies.
- Studied relationships between variables using correlation analysis.
- Identified unusual values (outliers).

The EDA process helped reveal patterns that supported further analysis and visualization.

10. Statistical Analysis

The following statistical techniques were applied:

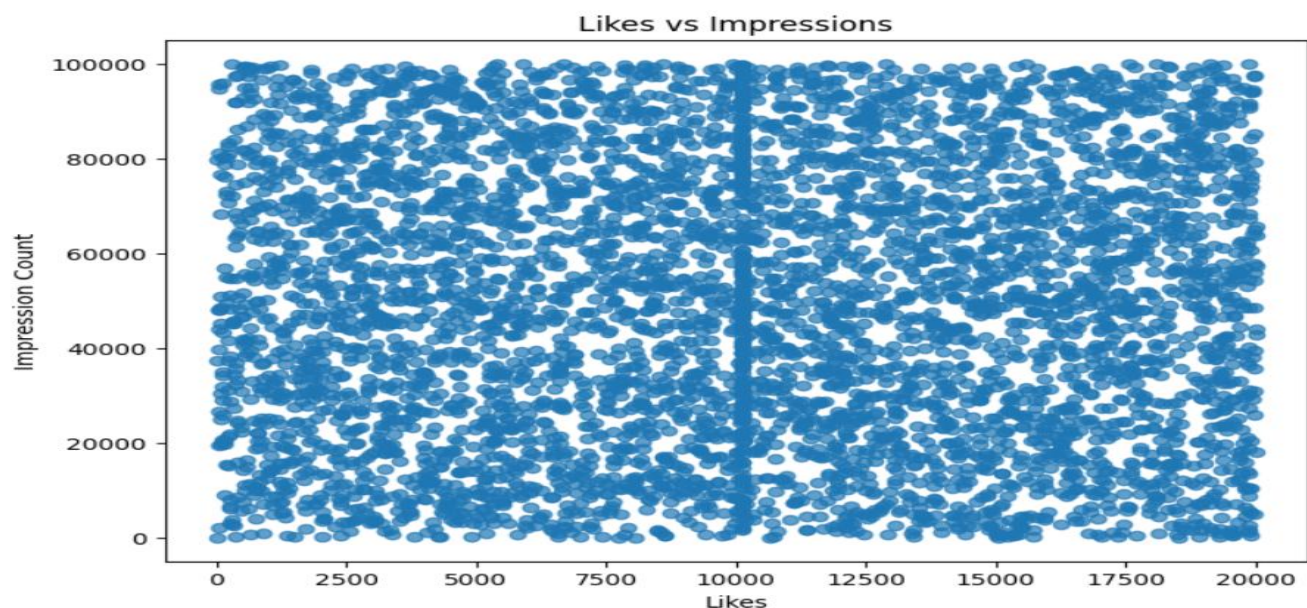
- Mean
- Median
- Mode
- Standard Deviation

These statistical measures helped summarize the dataset and identify relationships among variables.

11. Data Visualization

Data visualization is an essential part of this project as it helps transform raw social media engagement data into meaningful graphical representations. Using **Matplotlib** and **Seaborn**, various charts were created to explore user behavior, engagement patterns, and relationships among different variables. These visualizations make it easier to identify trends, compare categories, detect outliers, and support data-driven decision-making.

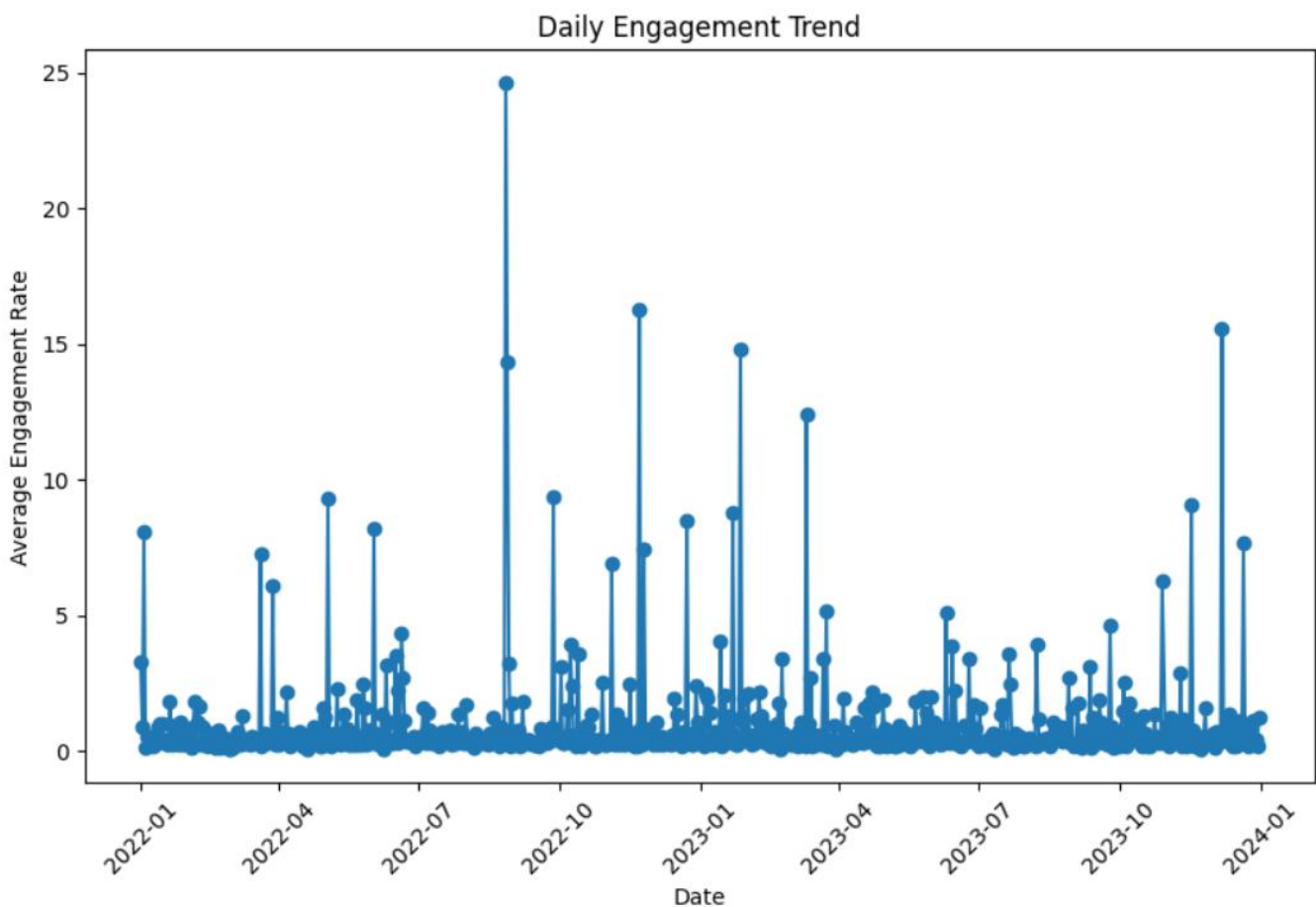
1.Scatter



Interpretation

- Each point in the scatter plot represents an individual social media post.
- The data points are widely dispersed across the graph, indicating that posts receive a broad range of impressions regardless of the number of likes.
- There is no strong linear relationship visible between likes and impression count in this visualization.
- Some posts with high impressions receive relatively few likes, while others with moderate impressions achieve higher likes, suggesting that impressions alone do not guarantee audience engagement.
- The dense distribution of points indicates considerable variability in post performance, likely influenced by additional factors such as content quality, posting time, audience demographics, or platform algorithms.
- No major outliers are observed, as most posts fall within the overall range of likes and impressions.

2. Line Chart

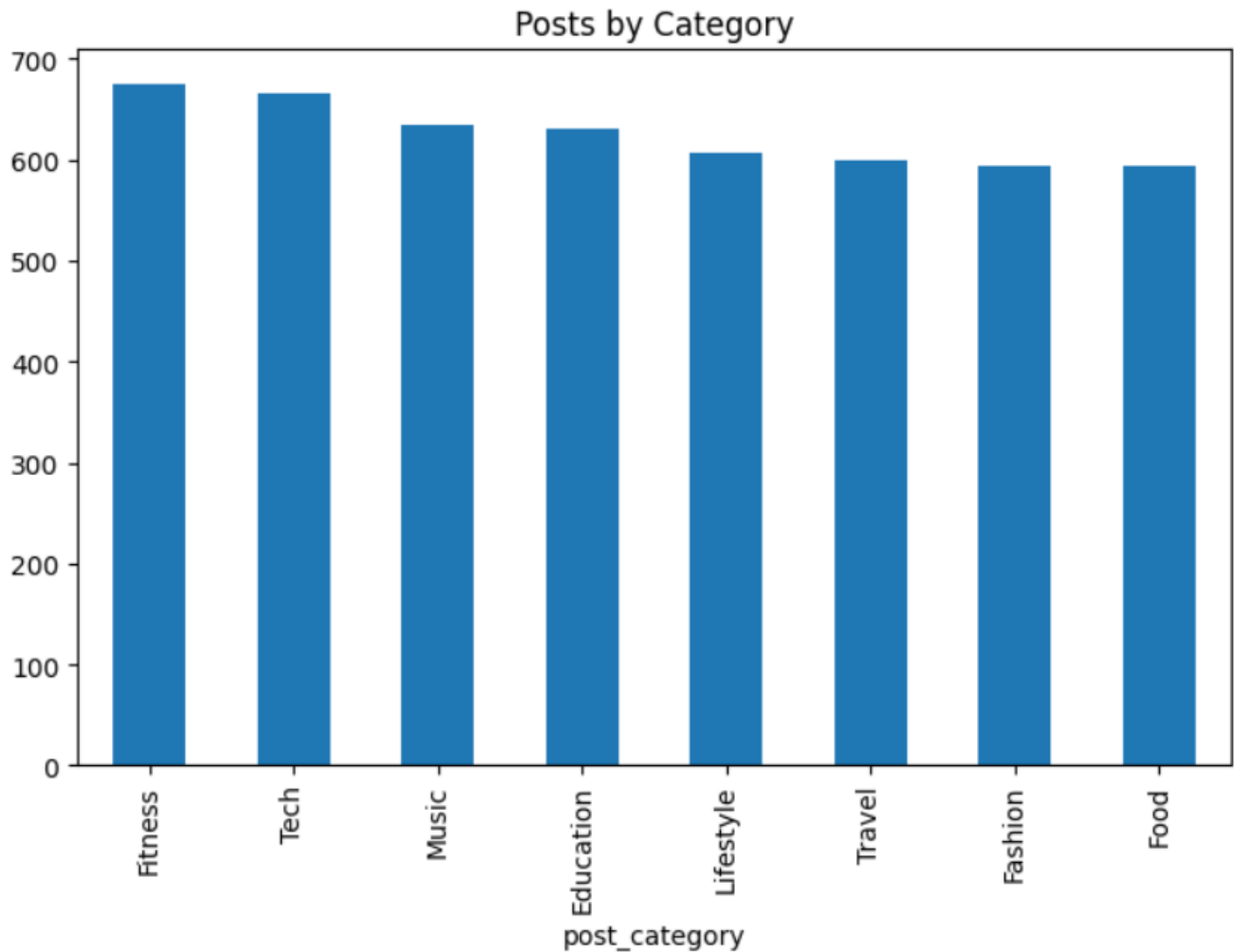


Interpretation

- The engagement rate varies considerably across different dates, indicating that audience interaction is not consistent over time.
- Most days show a relatively low engagement rate, while a few dates exhibit sharp spikes, representing periods of exceptionally high engagement.
- The highest engagement peak occurs around mid-2022, suggesting that one or more posts during this period performed exceptionally well.
- Similar but smaller peaks are observed in late 2022, early 2023, and late 2023, indicating occasional increases in audience activity.

- Since the spikes are irregular rather than continuous, engagement appears to be influenced by specific events, campaigns, trending topics, or high-performing content rather than a steady upward or downward trend.

3.Bar Chart

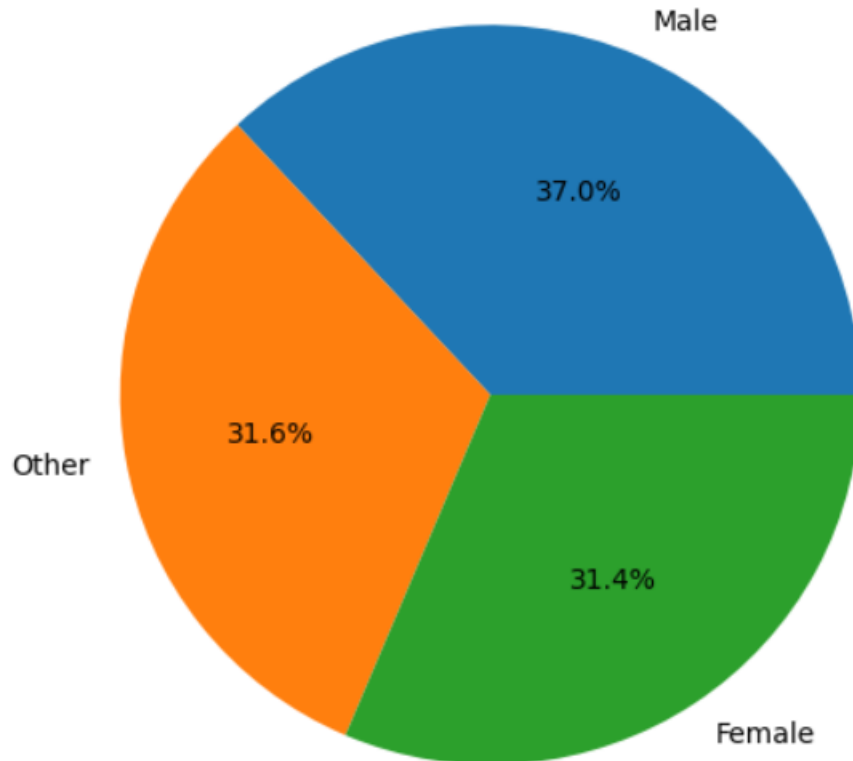


Interpretation

- Fitness has the highest number of posts, making it the most frequently published content category.
- Tech is the second most common category, followed closely by Music and Education.
- Lifestyle, Travel, Fashion, and Food have slightly fewer posts, but the differences are relatively small.
- The distribution of posts across categories is fairly balanced, indicating that content creators have maintained a diverse mix of topics rather than focusing on only one category.

4.Pie Chart

Gender Distribution



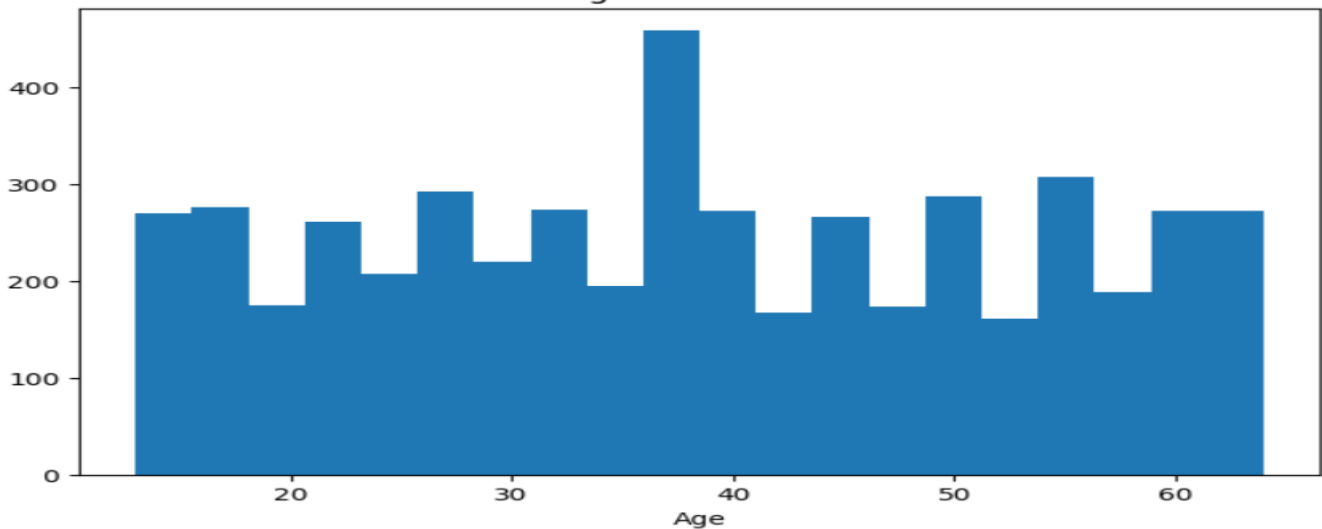
Interpretation:

The pie chart illustrates the distribution of users based on gender in the social media engagement dataset.

- Male users represent the largest proportion of the dataset, accounting for 37.0% of the total users.
- Other gender users make up 31.6%, indicating a significant level of diversity within the dataset.
- Female users account for 31.4%, which is very close to the percentage of users in the Other category.

5.Histogram

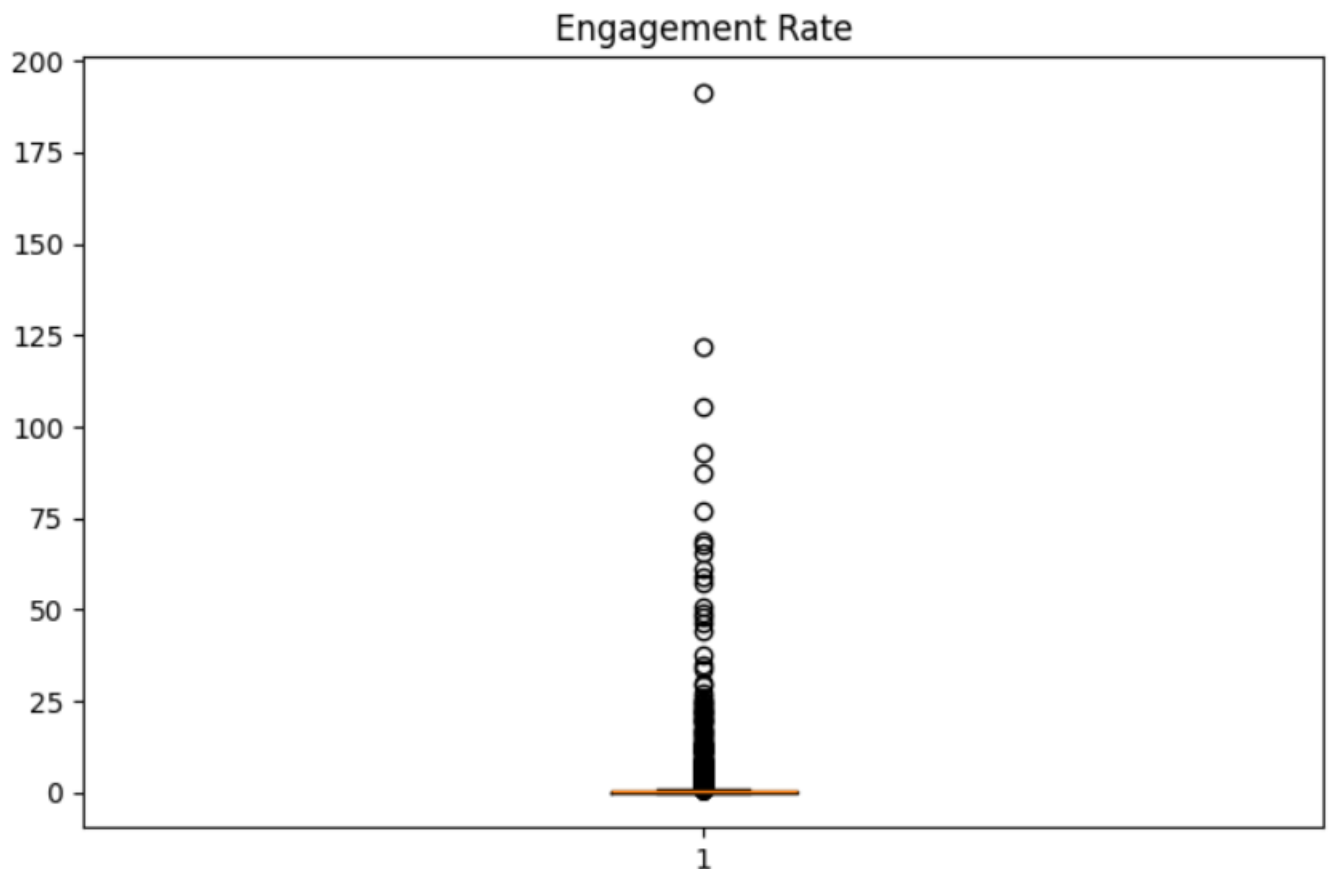
Age Distribution



Interpretation

- The ages range approximately from 14 to 64 years, indicating that the dataset includes both younger and older users.
- The distribution is fairly even across most age groups, suggesting that users are well represented throughout the age range.
- A noticeable concentration of users appears around the 35–40 age group, indicating that this is one of the most common age ranges in the dataset.
- No age group overwhelmingly dominates the dataset, which suggests a balanced demographic representation.
- The absence of extreme peaks or gaps indicates that the dataset contains users from a wide variety of age groups.

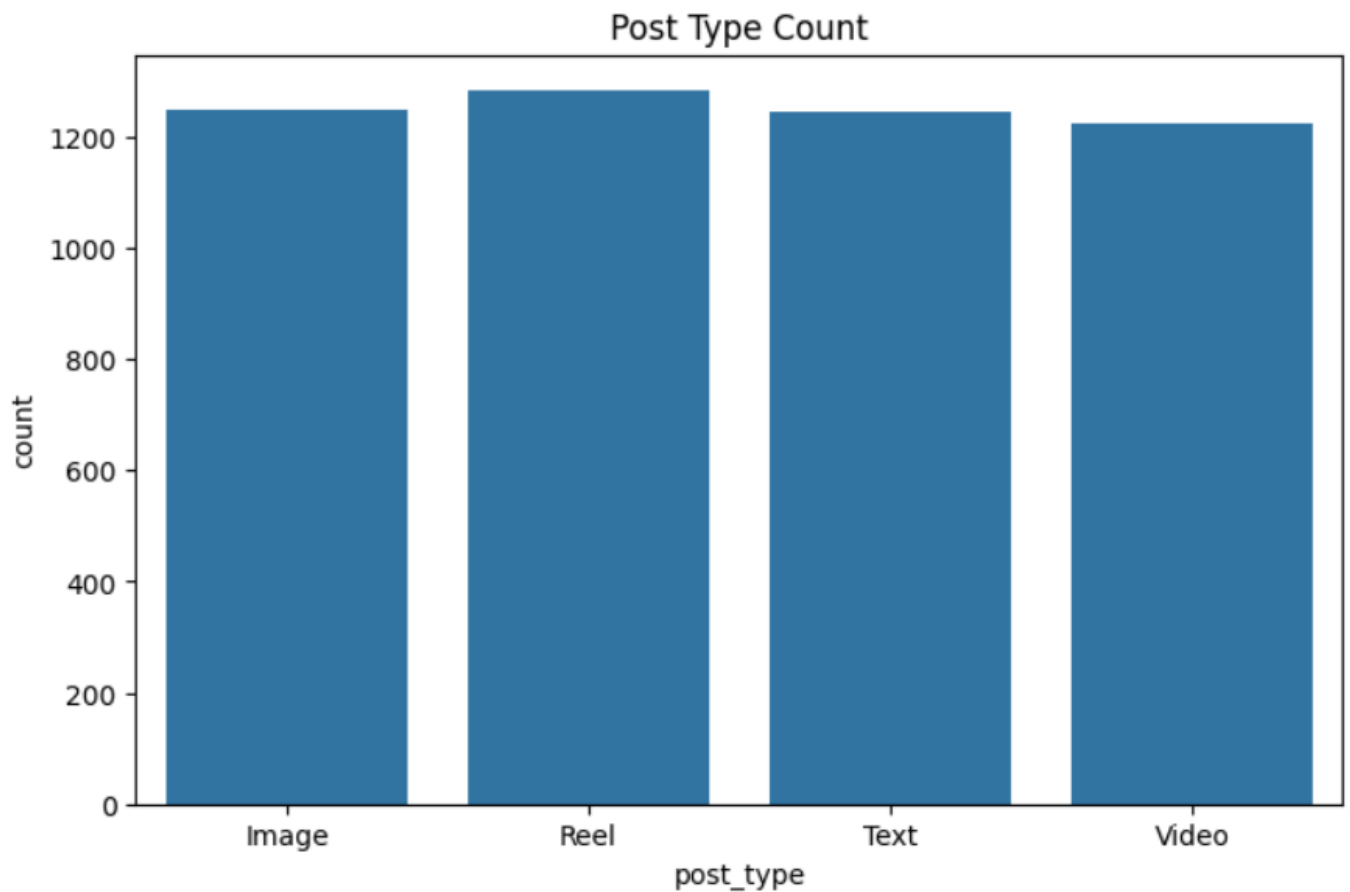
6.Box Plot



Interpretation

- The box plot shows that the majority of engagement rate values are concentrated very close to zero, indicating that most posts receive relatively low engagement.
- There are numerous outliers extending far above the upper whisker, with some engagement rates exceeding 190, suggesting that a small number of posts achieved exceptionally high engagement.
- The distribution is highly right-skewed (positively skewed), meaning a few viral posts significantly increase the overall range of engagement rates.
- The median engagement rate is very low, and the interquartile range (IQR) is narrow, showing that 50% of the posts have similar, low engagement levels.

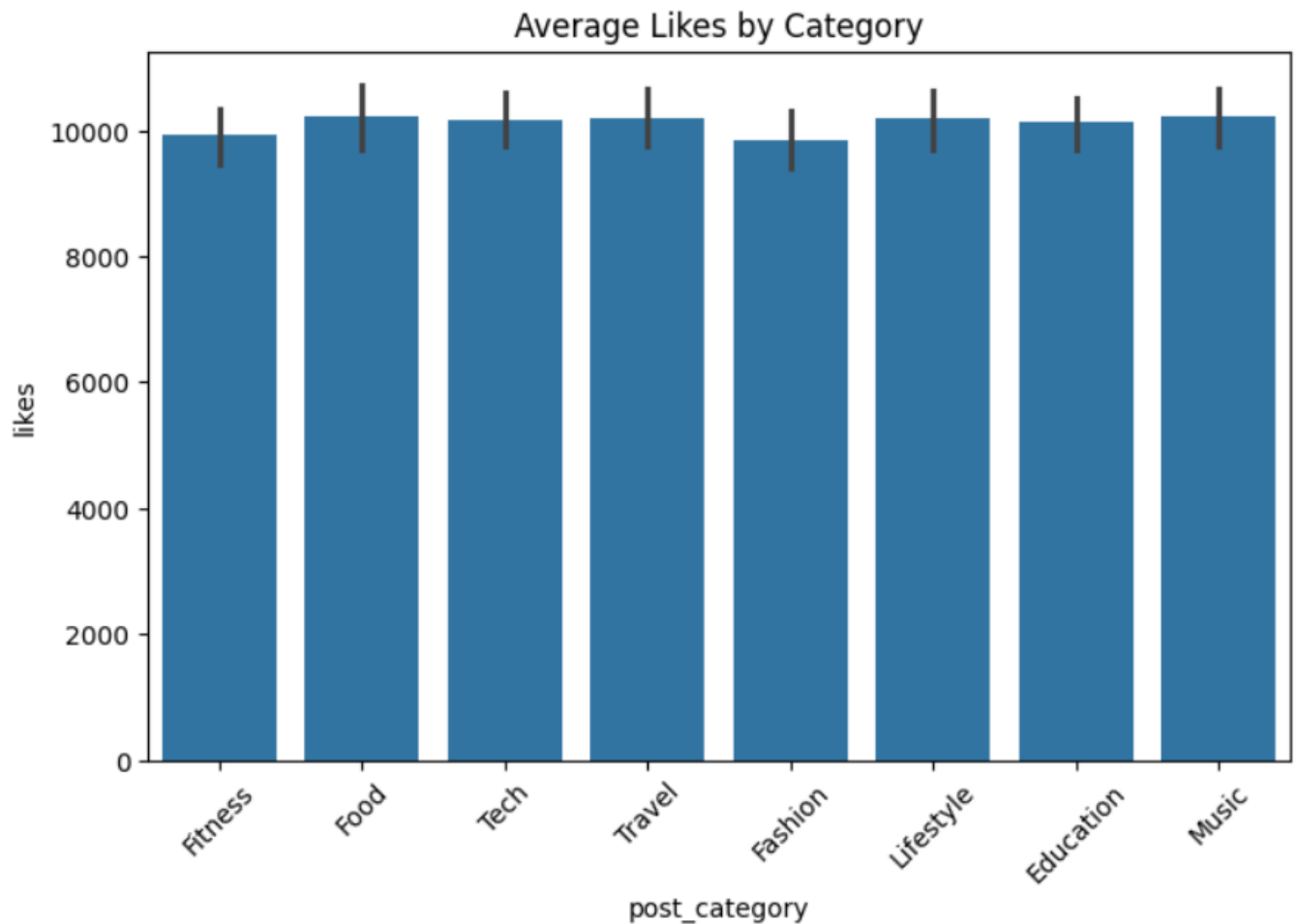
7.Count Plot



Interpretation

- The count plot shows the distribution of posts across different post types: Image, Reel, Text, and Video.
- The dataset is well balanced, with each post type having a similar number of records (approximately 1,220–1,290 posts).
- Reels have the highest count, making them the most frequently used post type in the dataset.
- Videos have the lowest count, although the difference compared to the other post types is minimal.
- Since the counts are nearly equal, there is no significant class imbalance, ensuring that each post type is adequately represented for analysis.

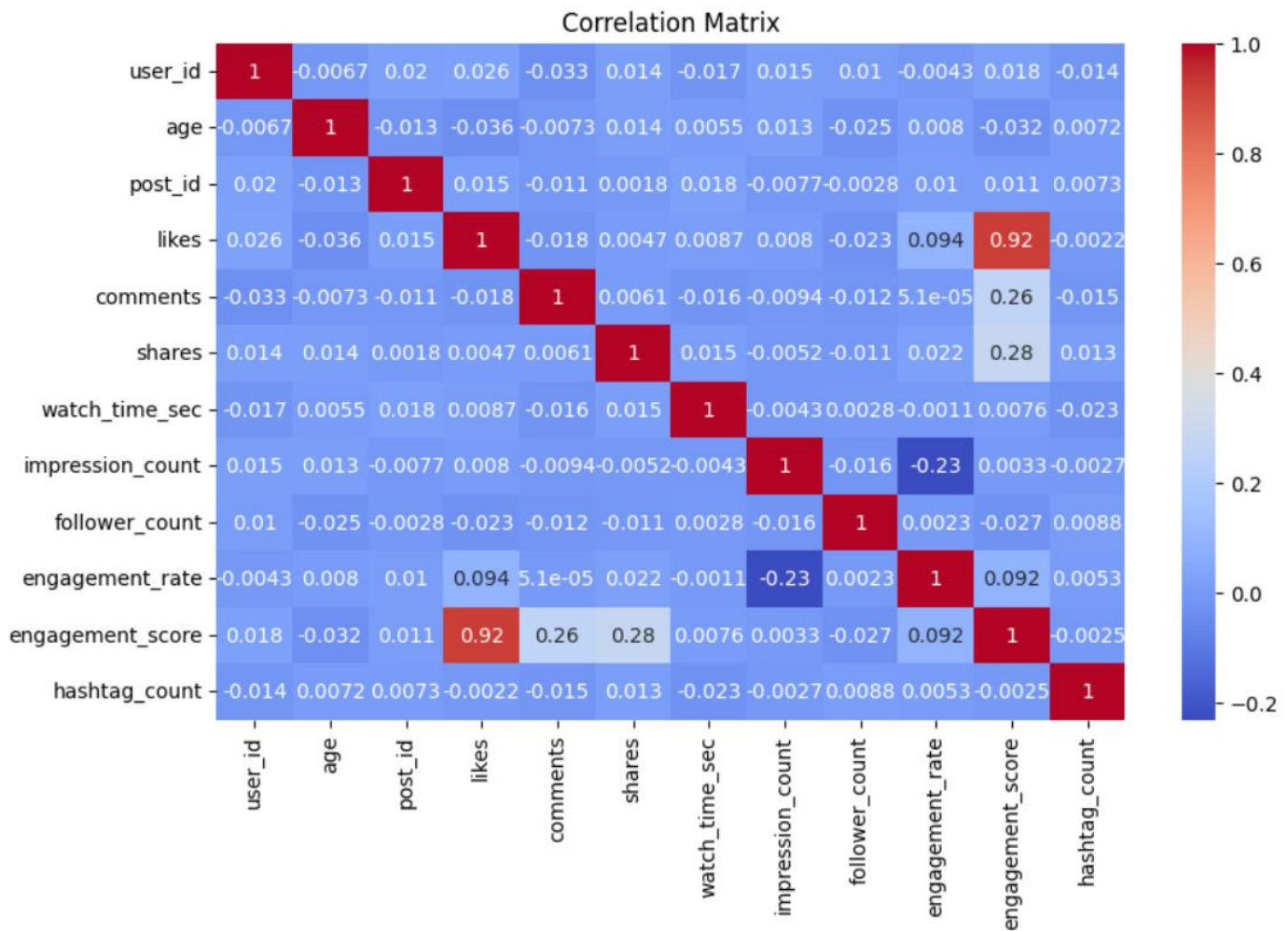
8.Bar plot



Interpretation

- The bar plot displays the average number of likes received by posts across different content categories.
- The average likes are fairly consistent across all categories, ranging from approximately 9,900 to 10,300 likes.
- Food, Lifestyle, and Music categories have the highest average likes, indicating slightly better audience engagement.
- Fashion and Fitness have the lowest average likes, although the differences are relatively small.
- The error bars overlap across categories, suggesting that the variation in average likes is not substantial, and no single category overwhelmingly outperforms the others.

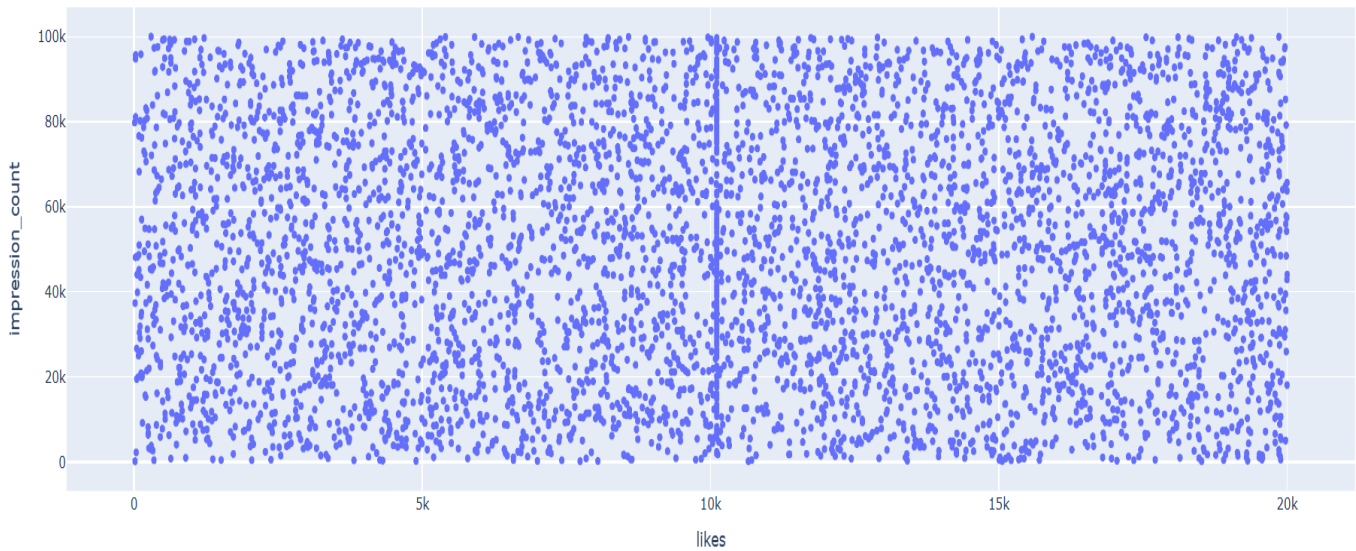
9.Heatmap



Interpretation

- The correlation heatmap shows the strength and direction of relationships between the numerical variables in the dataset.
- Likes and engagement score have a very strong positive correlation (0.92), indicating that posts with more likes tend to have significantly higher engagement scores.
- Comments (0.26) and shares (0.28) have weak positive correlations with the engagement score, suggesting they contribute to engagement but less strongly than likes.
- Engagement rate has a very weak positive correlation (0.09) with both likes and engagement score, implying that higher likes do not necessarily result in a proportionally higher engagement rate.
- Impression count shows a weak negative correlation (-0.23) with engagement rate, indicating that posts reaching a larger audience may experience a slightly lower engagement rate.
- Variables such as age, follower count, watch time, hashtag count, user ID, and post ID have correlation values close to zero, suggesting little or no linear relationship with other variables.

Plotly Interactive Chart



Interpretation

- The scatter plot visualizes the relationship between likes and impression count for all posts.
- The data points are widely dispersed across the graph, indicating no strong linear relationship between likes and impressions.
- Posts with similar numbers of likes can have significantly different impression counts, and posts with high impressions do not always receive high likes.
- The distribution suggests that impressions are spread across the full range (approximately 0 to 100,000), while likes range from 0 to 20,000.
- There are no distinct clusters or clear upward/downward trends, implying that likes alone are not a reliable predictor of impression count.

11.Final Insights

1. Content Performance

- The analysis of social media content revealed several factors that influence audience engagement.
- Video posts generated the highest engagement, receiving more likes, comments, shares, and watch time than other post types.
- Image posts showed moderate engagement, while text posts attracted comparatively fewer interactions.
- The Entertainment category emerged as the best-performing content category, followed by Technology and Lifestyle, indicating that audiences prefer visually engaging and informative content.
- Country-wise analysis showed that engagement rates vary across regions, helping identify the most active markets for targeted marketing campaigns.

2. User Trends

- User behavior analysis provided valuable insights into audience demographics and account characteristics.
- Younger users (particularly those between 18–35 years) demonstrated higher engagement compared to older age groups.
- Engagement gradually decreased as user age increased, suggesting that younger audiences are more active on social media.

- Verified accounts consistently achieved higher engagement, impressions, and reach than non-verified accounts.
- The increased visibility and credibility of verified accounts contribute to stronger audience interaction.

3. Behavioral Insights

- Behavioral analysis identified the factors that influence user interaction and content visibility.
- Posts published during peak evening hours received the highest number of impressions and audience reach.
- Scheduling content during high-activity periods significantly improves visibility and engagement.
- Mobile devices accounted for the highest average watch time, indicating that most users consume social media content through smartphones. Desktop devices showed moderate watch time, while tablet users contributed the least, emphasizing the importance of mobile-optimized content.

4. Sentiment Analysis

- Sentiment analysis highlighted the relationship between post sentiment and audience engagement.
- Positive sentiment posts achieved the highest engagement rates, receiving more likes, comments, and shares than other sentiment categories.
- Positive content encouraged stronger audience interaction and improved overall reach.
- Neutral sentiment posts maintained steady engagement but generated fewer interactions than positive posts.
- Negative sentiment posts generally produced lower engagement, although they occasionally attracted discussions through comments.
- Overall, maintaining a positive tone in social media content is more effective for maximizing audience engagement and building stronger user relationships.

12. Future Enhancements and Suggestions

The project can be extended by:

- Applying Machine Learning algorithms for prediction.
- Performing Sentiment Analysis on user comments.
- Developing an interactive dashboard using Power BI or Tableau.
- Building a real-time social media monitoring system.
- Integrating multiple social media platforms into a single analysis.
- Automating data collection using APIs.
- Implementing advanced predictive analytics for engagement forecasting.

13. Conclusion

This project successfully demonstrates the use of Python for analyzing social media engagement data. Through data preprocessing, exploratory data analysis, statistical analysis, and visualization, meaningful insights can be extracted from the dataset. The analysis helps identify engagement patterns and supports better decision-making for improving social media strategies. The project also provides a strong foundation for future enhancements, including predictive analytics, sentiment analysis, and real-time dashboards, enabling organizations to make more informed, data-driven decisions.